

ECS 251 Discussion: How to Give a Talk

1/5
ECS 251
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Q/A Opportunity

Any questions on course format?

Course policies?

Readings?

etc.?

How to give a talk

Slides and advice adapted from Thorsten Hoeffler (ETH Zurich) and Simon Peter (UWashington)

How to present a research paper

1. Read the paper (how to read a paper from class intro)
2. Present the paper (this discussion section)

How to draw an owl

1.



1. Draw some circles

2.



2. Draw the rest of the owl

Talking about research

- A good researcher can express their knowledge well
 - This assumes we have that knowledge!
 - So only advance to this step once you understand the paper :)
- Why is talking about research useful?
 - Order your thoughts, think about how to explain them
 - Communication with other researchers (your classmates!)
 - Gather feedback
 - Establish relationships
 - Eventually build a career

Why do you care?

- Presenting will be important for your career!
 - This is a good way to convince people to:
 - Give you good grades
 - Give you a job/promotion
 - Give you money/resources
 - Think you're smart
 - Like you, recommend you
- But presentation skills are hard to acquire
 - No one is good at this right away
 - Practice practice practice!

A good research talk

- Is centered around the audience (not you)
 - **Teaches**, engages, provokes, excites listeners
- Provides intuitions to the audience
 - Take away messages, surprises, “wow” effects
- It does not need to
 - Tell every detail (not possible anyway in the time limit)
 - Show off how smart you are

Tip: focus on clearly defined goals

- Pick your goals carefully
 - What do you want to communicate? What should people remember?

Anatomy of a talk

- Motivation, placement
 - 20%
- Key ideas
 - 70-80%
- Evaluation results
 - 0-10%
- Do not present results without an explanation
 - Don't just say “the authors made Application A run 50% faster”
 - Instead say “the authors present the FOO method relying on intuition BAR and that achieves 50% improvement for Application A”

The beginning of your talk

- You have about 2 minutes before your audience dozes off or starts reading email
 - Use them! Make every second count
 - Good approach: Present an abstract of your talk (or an elevator pitch):
 - Problem / motivation
 - Approach / idea
 - Experiments / results
 - Broader meaning / impact
- Answer these questions within two minutes:
 - What is the problem?
 - Why is this talk interesting? Why should I listen?

Communicating the key idea

- Pick a goal for the talk
 - Plan and make key points in your head. Organize the whole talk around these key points. Pick no more than three (better: one)
 - Be explicit, very explicit: “If you remember nothing else from this talk, remember this”
 - Repeat, repeat (but don’t be annoying)
- Do NOT be shallow, be deep
 - Avoid overviews
 - Do NOT ramble
 - Get to the meat quickly (assume we’ve all read the paper already)

Examples

- Are your main weapon
- Make your own examples, try to avoid just using the paper's
- Ideally have a motivating example at the beginning
 - Maybe pose a question to get the audience thinking:
 - e.g., "What is the maximum speedup if we can solve this problem?"
- Illustrate the idea in action
 - From different perspectives
 - Show corner cases, highlight shortcomings
- Images say more than 1000 words!

What to use

- Enthusiasm
 - Be excited, pull the audience with you
 - Don't be afraid to move around, but don't pace or fidget
- Your brain!
 - Review/polish your slides before the talk
 - Have the storyline down
 - Focus on the key ideas
- Animations and graphics
 - Can be helpful (aesthetically as well as informative)
 - Nice slides make people more receptive

What to omit

- Do not present excessive related work
 - You can mention it in your slides, or have backup slides
 - Give credit
- Do not present too many technicalities
 - Audience probably won't follow
 - Put details in the backup slides in case you do get a question on them
- Do not exaggerate with animations
 - Animations are good, but too many are difficult to follow
- Do not clutter your slides

How to present

- Be (or at least appear) confident
 - Don't forget to breathe! :)
- Make eye contact
 - Look around, don't stare at a single person
 - Tip: identify a nodder (these people always exist). They'll give you confirmation
- Watch the audience
 - Sometimes they ask questions, don't let them interrupt you but serve their questions
 - Questions are great, ask some and answer them!
- Finish on time!
 - Skip slides if necessary, never ask "should I continue?"
 - No polite person would ever say, "no, thanks"

Miscellaneous

- Standard stuff
 - Avoid errors no slides
 - Face the audience
 - Check your laptop before
- Practice, practice, practice
 - Give your talk a couple of times before you present it publicly
 - Present to your group mates, me during office hours, your friends, your cat...
- You'll attend more talks here than you'll give
 - Engage, help your fellow students!
 - Ask questions, participate in the discussion, be awake

Miscellaneous

- You may borrow slides
 - But, be aware of the context of the slides
 - E.g., conference slides are intended for a very specialized audience
 - May have to adapt the slides for a more general audience (and the time frame)
 - And give credit
- Timing and cadence
 - Between 1-2 minutes per slide is common
 - Try not to rush too quickly through slides or spend too much time on a slide

Suggested lecture structure

- 5 minutes on background
 - What problem is this paper trying to solve? What other solutions have been attempted? Why do they fall short? Any historical context?
- 10 minutes on core concepts
 - What is this paper's solution to the problem? What is their core insight or design? Why is this paper important?
- 10 minutes for small group discussion
 - Your group and I will walk around the groups and help answer questions
- 15 minutes whole class discussion
 - Your group will help lead a full class discussion, perhaps based on questions from groups

Quiz questions to go with the lecture

- You will develop 4 quiz questions based on the reading
 - Should be on the important concepts or ideas from the paper, not minute details
 - E.g., **don't** ask “How much faster was Exokernel than Ultrix for null procedure calls?”
 - **Do** ask “What are secure bindings in Exokernel?”
- Can be definition-based questions (“What is <thing>?”) or more open ended questions (“Why did the authors go with design decision A?” or “Compare/contrast” questions)
- We may use your quiz questions when developing the weekly quizzes!
 - Your opportunity for easy points! ;)

How your presentation will be graded (lead a lecture)

- Clarity of background (3 pts)
- Clarity of main concept (6 pts)
- Smoothness and polish of presentation (2 pts)
- Question and answer leading (2 pts)
- Turning in slides by the class before your lecture (3 pts)
- Turning in 4 potential quiz questions (4 pts)

How your presentation will be graded (final project)

- Clarity and quality of presentation (2 pts)
- Explanation of problem and context (3 pts)
- Explanation of the system you've built (6 pts)
- Handling of the Q/A session (3 pts)